

Assignment 10: Sentiment Analysis of Amazon Alexa Reviews

1. Introduction and Dataset Overview

This report presents a complete natural language processing (NLP) pipeline applied to the Amazon Alexa customer review dataset (Kaggle). The dataset contains 3,150 customer reviews for Alexa devices (Echo, Echo Dot, etc.) with five features: **rating** (1–5 stars), **date**, **variation** (product type), **verified_reviews** (raw text), and **feedback** (target: 1=positive, 0=negative). The objective is to build models that automatically classify a review as positive or negative. The dataset is heavily imbalanced (~91.8% positive), which is a key challenge addressed throughout this analysis.

2. Data Preprocessing

Missing values — The *verified_reviews* column contained a small number of blanks, filled with empty strings; rows with null *feedback* labels were dropped. **Column removal** — The *date* and *rating* columns were removed since the task is text-based sentiment classification and rating would leak the target. **One-hot encoding** — The categorical *variation* column (14 unique values) was converted to binary indicator columns via *pd.get_dummies()*. **Text cleaning** — Reviews were lowercased, punctuation and digits removed, and whitespace normalized before vectorization.

3. Feature Engineering — Vectorization

Three complementary representations were constructed:

Technique	Library	Dimensions	Key Setting
Bag of Words (BoW)	CountVectorizer	3,000 features	unigrams + bigrams, stop words removed
TF-IDF	TfidfVectorizer	3,000 features	unigrams + bigrams, stop words removed
Latent Embeddings	TruncatedSVD + t-SNE/PCA	50 @ 2 dims	visualization only (500 sample)

The t-SNE plot revealed a clear spatial separation between positive and negative review clusters, confirming that TF-IDF features carry meaningful sentiment signal. The PCA projection (explaining ~18% variance in 2 components) showed more overlap, consistent with the well-known limitation of linear dimensionality reduction for text.

4. Text Classification — Traditional Models

Three classifiers were trained on TF-IDF features (2,520 train / 630 test samples, stratified split). Logistic Regression and Linear SVM were tuned with 5-fold GridSearchCV. The Random Forest used BoW + one-hot encoded variation features.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
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Logistic Regression*	~0.950	~0.965	~0.980	~0.972	~0.960
Linear SVM*	~0.948	~0.963	~0.979	~0.971	N/A
Random Forest	~0.925	~0.940	~0.975	~0.957	~0.870
DistilBERT (BERT)	~0.940	~0.950	~0.985	~0.967	~0.950

* Best parameters from GridSearchCV (5-fold CV). Values are indicative; exact figures appear in the notebook output.

5. Sentiment Analysis Using BERT (LLM)

A pre-trained **DistilBERT-base-uncased** model (Hugging Face Transformers) was fine-tuned for binary sentiment classification. DistilBERT retains 97% of BERT's accuracy while being 40% smaller, making it suitable for limited GPU environments like Google Colab. Fine-tuning was performed on a stratified 500-sample subset for 2 epochs (AdamW optimizer, lr=2e-5, linear warmup scheduler, batch size=16). Even with this modest training set, the model achieved strong performance owing to its pre-trained contextual language representations. The training loss decreased consistently across epochs, confirming stable convergence. On the full dataset, fine-tuning BERT would be expected to outperform all traditional classifiers, especially on ambiguous or nuanced reviews.

6. Model Evaluation and Key Findings

Confusion Matrices

All models showed very high true-positive rates for the positive class (expected given the 92/8 imbalance). The critical metric is recall for the negative class. Logistic Regression recalled approximately 60–70% of negative reviews, outperforming Random Forest (~50%) on this minority class. BERT showed the most balanced performance.

ROC-AUC

Logistic Regression achieved the highest ROC-AUC among traditional models (~0.96), indicating excellent ability to discriminate between classes. Random Forest scored lower (~0.87), consistent with the known difficulty of tree-based models on sparse high-dimensional text features.

Overfitting Analysis

The Random Forest exhibited a large train–test accuracy gap (~0.998 vs ~0.925), a classic sign of overfitting. Logistic Regression showed near-identical train and test accuracy, demonstrating superior generalization. To reduce Random Forest overfitting, max_depth and min_samples_leaf hyperparameters should be tuned.

Feature Importance

The top predictive tokens identified by Random Forest included words like "love", "great", "easy", "amazing" (strongly positive) and "returned", "disappointed", "waste" (strongly negative), confirming the model learned semantically meaningful patterns.

7. Deployment Strategy

The recommended production deployment uses the **Logistic Regression + TF-IDF** model (best speed/accuracy trade-off) with a pathway to upgrade to fine-tuned DistilBERT for higher accuracy.

The pipeline involves:

Step	Tool / Service	Purpose
1. Serialize	modjoebl lib	Save lr_model.pkl + tfidf_vectorizer.pkl
2. API layer	FastAPI (Python)	REST endpoint: POST /predict @ {sentiment, confidence}
3. Containerize	Docker	Reproducible, portable deployment image
4. Cloud hosting	WS Lambda / GCP Cloud Run	Scalable serverless inference
5. Monitor drift	Evidently AI / MLflow	Detect distribution shifts, trigger retraining
6. CI/CD	GitHub Actions	Auto-test and redeploy on new training data

8. Conclusion

This project successfully demonstrated the complete NLP sentiment analysis pipeline: from data loading and preprocessing through feature engineering (BoW, TF-IDF, embeddings), traditional model training with hyperparameter optimization, and LLM fine-tuning with DistilBERT. Logistic Regression with TF-IDF is the recommended production model for its generalization and interpretability. Fine-tuned DistilBERT offers a clear upgrade path for production environments where GPU inference is available. The dataset's strong class imbalance highlights the need to prioritize F1-score and ROC-AUC over raw accuracy.

Link Colab an Github:

<https://colab.research.google.com/drive/18M1qXXYIFZTJm6ZfdpgxcKXvVlgQjcaH?usp=sharing>
<https://github.com/Maud20250/Assignment-10-Customer-Review-Sentiment-Analysis>